

Denver Water

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Water Quality and Treatment Manager



Quick Overview

- Introduction to Denver Water
- Challenges and Opportunities in the Water Sector
- Design Storm Challenge
 - Data sources
 - Scenarios

About Us

1.5M

people served

25%

of Colorado population

4,000

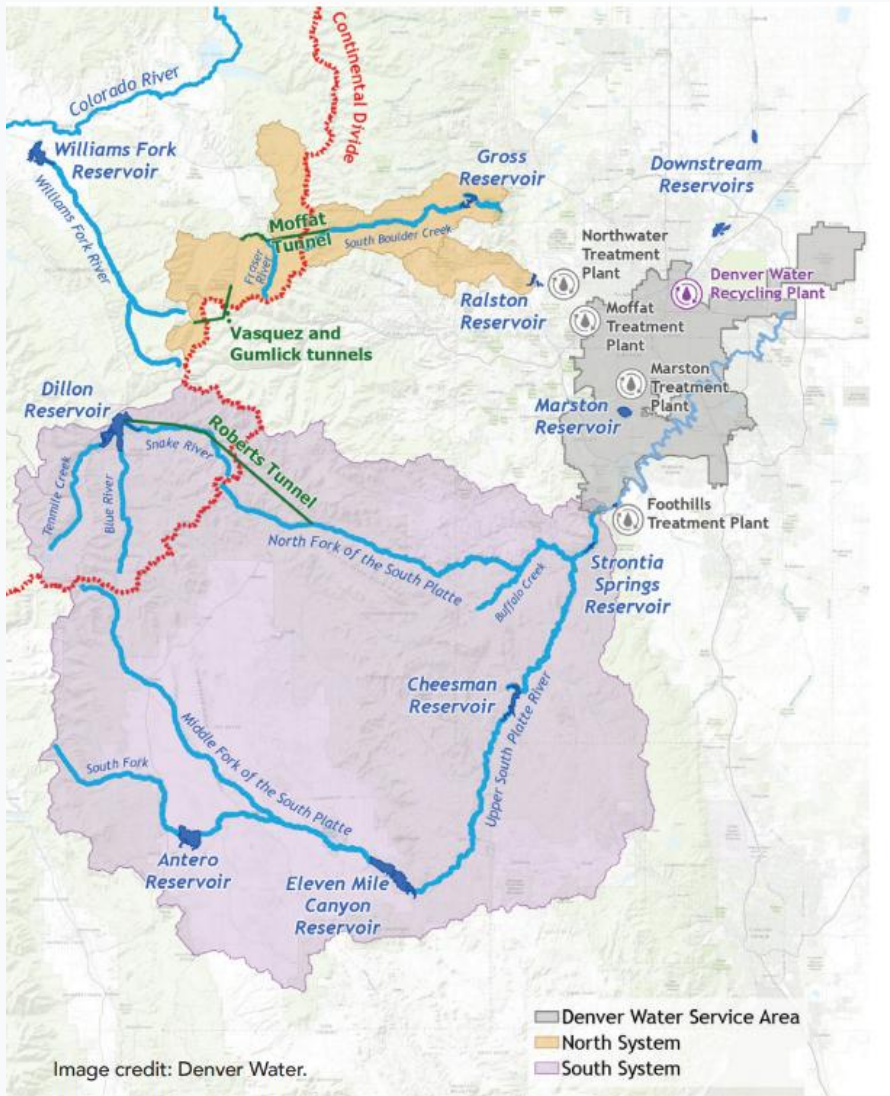
sq. miles collection

3,000

miles of pipelines

- Colorado's oldest and largest water utility, formally became Denver Water in 1918
- Proudly serve 1.5 million people across the Denver metro area
- 25% of the state's population, <2% of all water used across the state
- Collection system spans about 4000 square miles
- Distribution system has 3000 miles of water pipelines

About Denver Water's System



Two systems. One integrated supply.

- Two Systems, North and South
- 80% South System
- 20% North System
- Snowpack, surface water
- Own and operate 9 reservoirs across the system

COLLECTION › TREATMENT › DISTRIBUTION

Challenges and Opportunities - Water Sector

Challenges

- Water Quantity issues in the arid west.
 - 2026 drought
- Water Quality challenges, environmental and human impact.
 - Wildfires, drought, urbanization, agriculture and industrial use
- Federal and State regulation change and adaptability
- Resiliency of our treatment plants and system (changing water sources and quality, aging infrastructure, and cost)

Changing landscape

- New technologies
- Ability to access, compile and apply data (internal and national datasets)
- Proactive responses and planning

It's a drought.

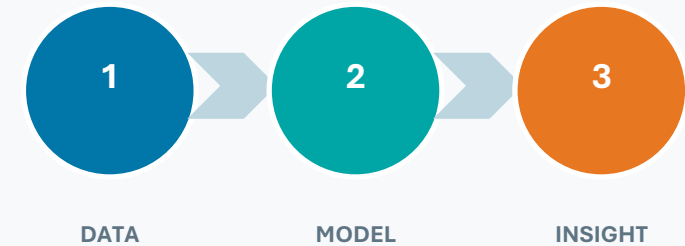
USE ONLY WHAT YOU NEED.SM



Design Storm

Use your backgrounds and skillsets to develop creative solutions to the scenarios presented to you today.

The scenarios and datasets provided are used every day across our utility to give predictions, model scenarios and give insight on what's occurring in our collection system.



Data Sources

National Datasets

- Snowpack
- precipitation
- weather
- streamflow
- water quality parameters

Denver Water Public data

- Total Organic Carbon(TOC) and Alkalinity data from grab samples
- Continuous reservoir data, depth and physical parameters

SCENARIOS

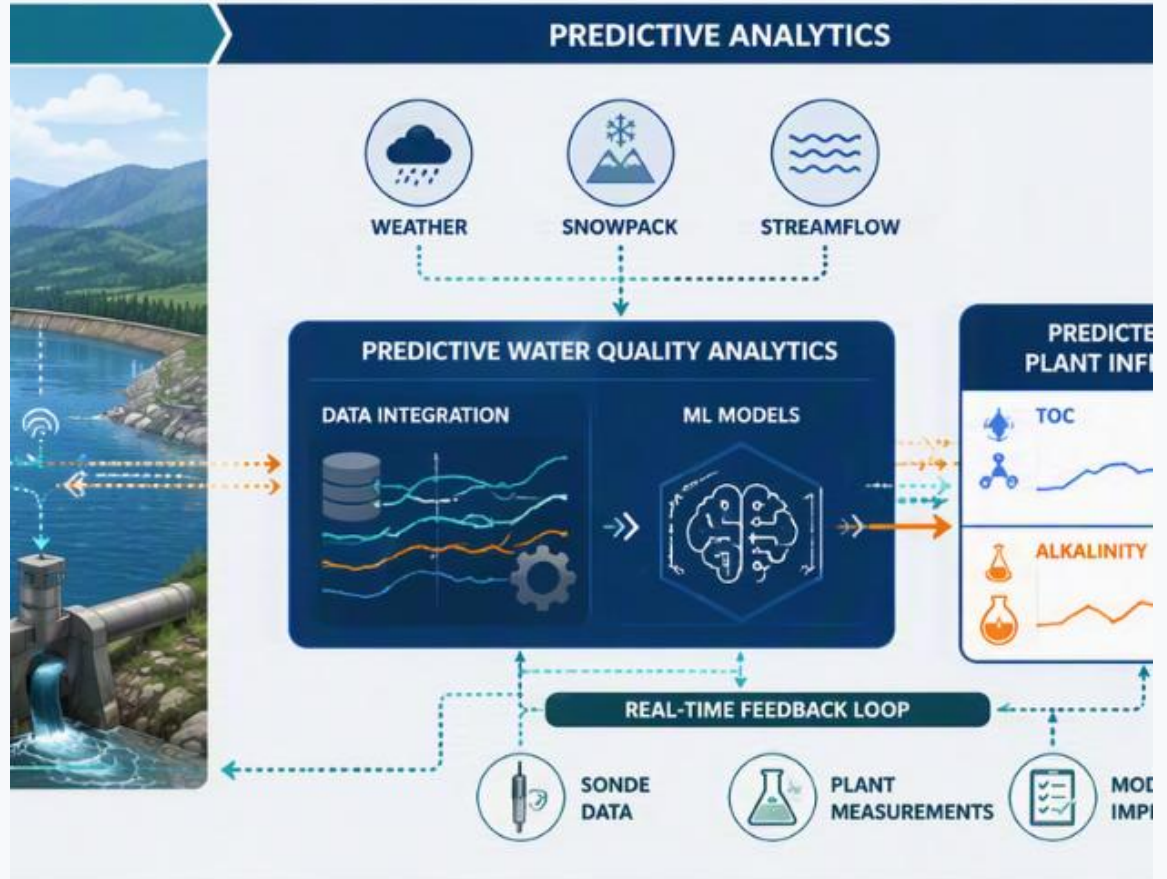
Scenarios to Consider

Three applied challenges using real utility data.

TOC and Alkalinity Predictive Model

Can watershed, hydrologic, and reservoir monitoring data provide enough advance warning to accurately predict TOC and alkalinity arriving at Foothills and give treatment staff actionable time to prepare?

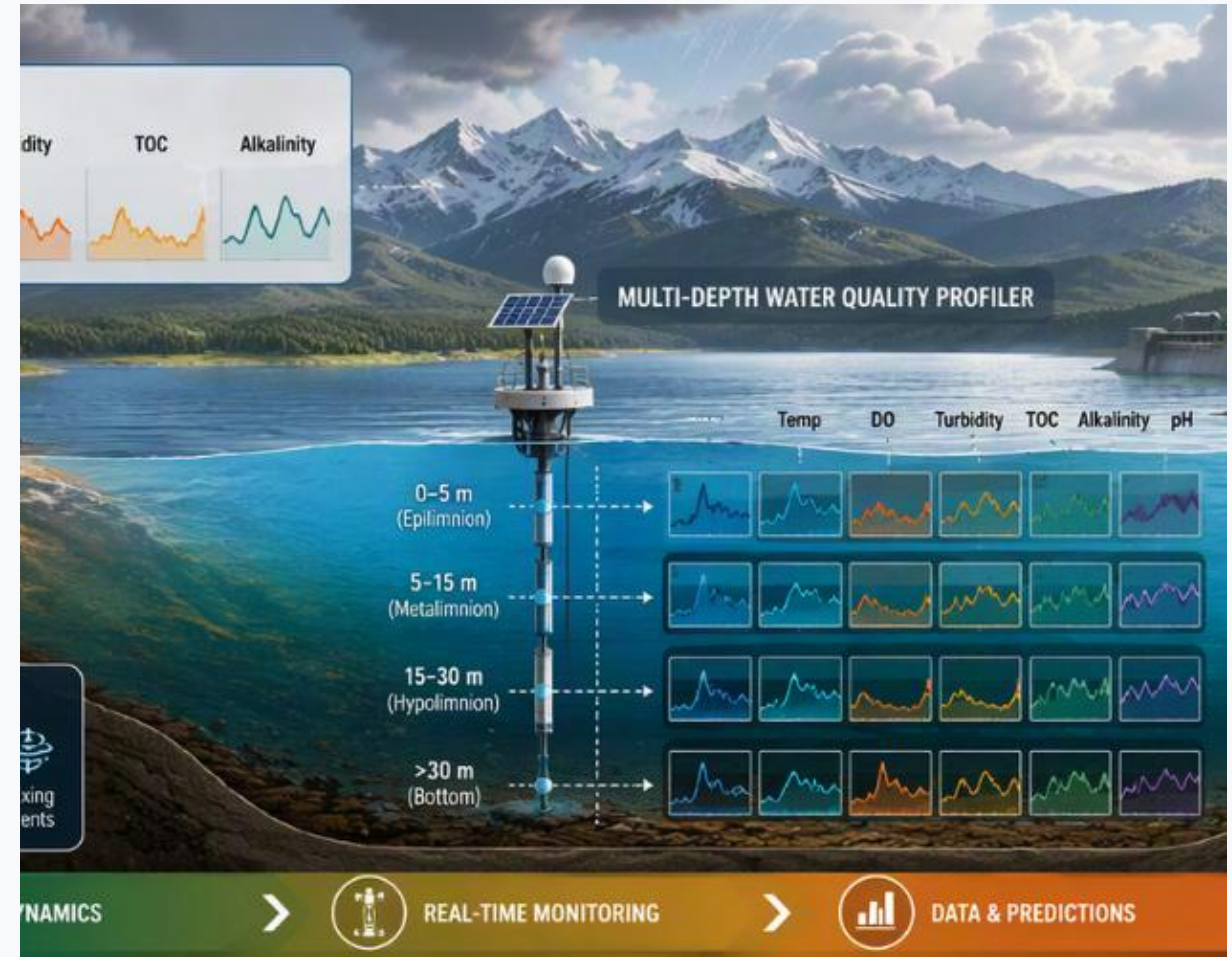
- Using national datasets upstream of Strontia Springs Reservoir; predict TOC and Alkalinity concentrations a few days ahead hitting Foothills treatment plant.
- ★ Introduce real-time Strontia profiling sonde data for improving predictions. This location is much closer to the Foothills influent which may influence accuracy but also impact lead/lag time
- Machine learning aficionados could try different models (like support-vector machines), lag-times, and feature engineering to see if they can improve performance.
- Develop a web-application for viewing all the relevant data for predictions (streamflow, weather, USGS sonde measured water quality), and projected TOC and/or alkalinity.



Storm and Runoff events and Real Time Data Application

Given current watershed and reservoir conditions, how is an incoming storm or runoff event likely to affect source-water quality, when will that impact arrive, and what conditions might we expect at different depths within Strontia Springs Reservoir?

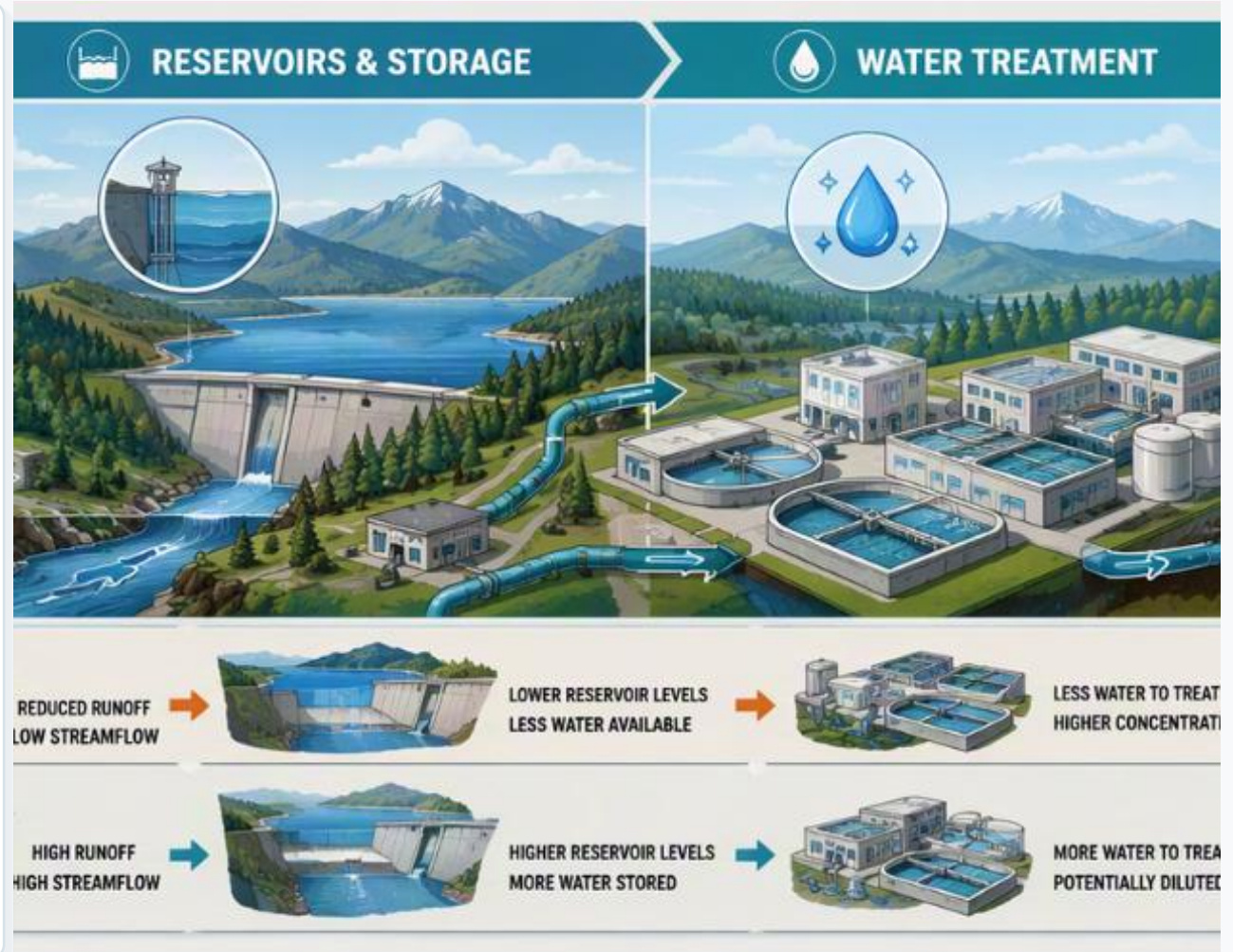
- Model precipitation events impact on real time water quality parameters above Strontia Springs reservoir
- Further adapt that model to evaluate Strontia Springs Reservoir Sonde data, how do water quality parameters change and distribute by depth in the reservoir and how is that impacted by storms and spring runoff
- Look at historical datasets and model how major precipitation events have impacted water quality in the system



Snowpack and Surface Water System Function

Develop an interactive model that visualizes how water and water-quality conditions move from the watershed through the collection system to the treatment plants and evaluate how hydrologic and seasonal events influence that movement.

- Develop a model or visualization of datasets that demonstrate how water enters and moves through the system to our treatment plants.
- Focus on different aspects of the water system (ex) Historical snowpack and streamflow, and how those change from year to year (drought vs wet years)
- Choose a parameter or parameters to visualize and follow through the system, see if different factors have an impact on transport (snowpack, rainstorms, lake turnover)



Support and Questions along the way?

- Jake Slawson, Denver Water Data Scientist
 - 9/24: available 3-4pm,
 - 9/25: available throughout the morning for questions, and attendance at the report out
- Cassidi Rosenkrance, WQ&T Manager
 - 9/24: available 3-5pm, Q&A session at 4:45pm.
 - 9/25: available throughout the morning for questions, and attendance at the report out

We are excited to see what you develop and are here to support along the way!